

QUESTION-14 Create a System C-integrated Embedded C simulation for a multi-sensor data acquisition system.

Abstract

Embedded systems are widely used in industrial automation for acquiring and processing data from multiple sensors. This project presents a SystemC-integrated Embedded C simulation of a multi-sensor data acquisition system. The system collects temperature, pressure, and humidity data, processes the information using Embedded C, and models the hardware behavior using SystemC. The simulation demonstrates structured software design, real-time task scheduling, and efficient communication between hardware and software components. The proposed model provides a reliable platform for industrial monitoring applications and can be extended for Internet of Things (IoT) and smart manufacturing systems.

Keywords: Embedded C, SystemC, Data Acquisition, Sensors, Industrial Automation, Real-Time Systems.

Introduction

Embedded systems are specialized computer systems designed to perform dedicated tasks within larger systems. Modern industrial automation relies heavily on embedded systems to continuously monitor and control physical parameters such as temperature, pressure, humidity, vibration, and gas concentration. A multi-sensor data acquisition system collects information from different sensors simultaneously and processes it in real time. Embedded C is commonly used for programming microcontrollers due to its efficiency and direct hardware access. However, before implementing the software on actual hardware, it is beneficial to simulate the complete system using SystemC. SystemC is a C++ class library that supports system-level modeling and simulation. It enables engineers to model hardware components, timing behavior, and communication protocols. Integrating Embedded C with SystemC improves system verification and reduces development time before hardware implementation.

Problem Statement

Industrial environments require reliable and real-time monitoring of multiple physical parameters. Conventional systems may suffer from delayed processing, inefficient scheduling, and high development costs due to hardware testing. Therefore, a simulation-based approach

using Embedded C and SystemC is required to model, analyze, and validate the operation of a multi-sensor data acquisition system before deployment.

Objectives

- Design a multi-sensor data acquisition system.
- Read sensor values using Embedded C.
- Simulate the embedded system using SystemC.
- Implement real-time periodic sampling.
- Analyze system performance.
- Demonstrate industrial applications.

System Requirements

- SystemC Library
- GCC Compiler
- Visual Studio Code
- C++ Compiler

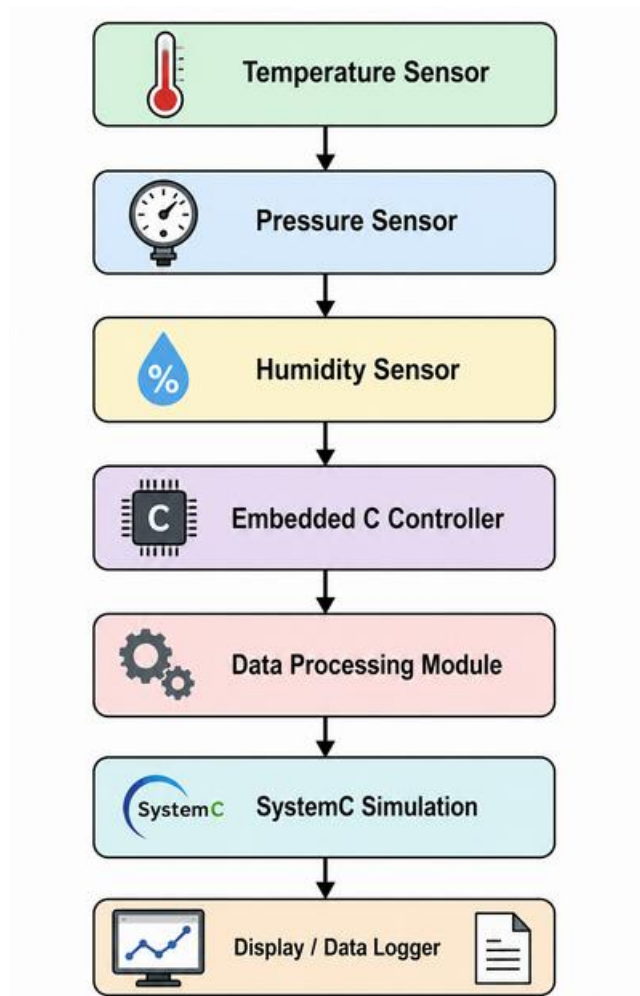
Methodology

The proposed system consists of three sensors connected to an embedded controller. Embedded C is used to acquire sensor data periodically. SystemC models the timing behavior of the controller and sensors.

The workflow includes:

- Sensor Initialization
- Data Acquisition
- Signal Processing
- Threshold Checking
- Data Display
- Continuous Monitoring

System Architecture



Algorithm

Step 1: Start the system.

Step 2: Initialize sensors.

Step 3: Read sensor values.

Step 4: Store sensor readings.

Step 5: Check threshold values.

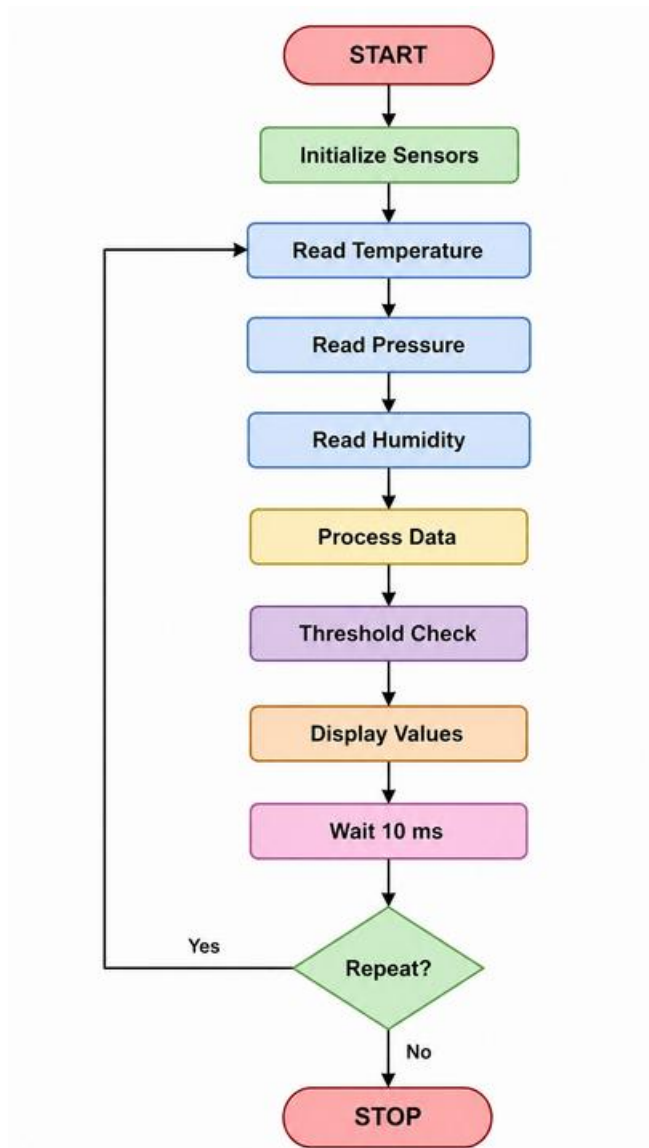
Step 6: Display sensor values.

Step 7: Wait for 10 ms.

Step 8: Repeat Steps 3–7 continuously.

Step 9: Stop.

Flowchart



SystemC Program:

main.c

```
1  #include <stdio.h>
2  int main()
3  {
4      int temp = 30;
5      int pressure = 1005;
6      int humidity = 65;
7      int time;
8      printf("Multi-Sensor Data Acquisition System\n");
9      printf("-----\n");
10     for(time = 0; time <= 50; time += 10)
11     {
12         printf("Time: %d ms\n", time);
13         printf("Temperature = %d C\n", temp);
14         printf("Pressure    = %d hPa\n", pressure);
15         printf("Humidity    = %d %%\n", humidity);
16         if(temp > 35)
17             printf("Warning: High Temperature!\n");
18         printf("-----\n");
19     }
20     return 0;
21 }
```

Output

Multi-Sensor Data Acquisition System

Time: 0 ms

Temperature = 30 C

Pressure = 1005 hPa

Humidity = 65 %

Time: 10 ms

Temperature = 30 C

Pressure = 1005 hPa

Humidity = 65 %

Time: 20 ms

Temperature = 30 C

Pressure = 1005 hPa

Humidity = 65 %

Time: 30 ms

Temperature = 30 C

Pressure = 1005 hPa

Humidity = 65 %

Advantages

- Supports multiple sensors.
- Real-time monitoring.
- Easy hardware-software integration.
- Reduces development cost.
- Easy debugging.
- High reliability.
- Scalable design.

Applications

- Smart factories
- Industrial automation

- Environmental monitoring
- Medical monitoring systems
- Automotive electronics
- Agriculture monitoring
- IoT systems

Result

The simulation successfully acquired data from multiple sensors and demonstrated periodic real-time monitoring using Embedded C and SystemC. The system processed sensor readings at fixed intervals, validated threshold conditions, and displayed the monitored parameters accurately.

Conclusion

A SystemC-integrated Embedded C simulation for a multi-sensor data acquisition system was successfully designed and analyzed. The implementation demonstrated structured programming, periodic real-time execution, and efficient handling of multiple sensor inputs. The simulation verified the functionality of the embedded application before hardware deployment, making it suitable for industrial automation, process monitoring, and IoT-based applications.